

Wajibu System Architecture

Turn Kenyan county tender & budget PDFs into plain-language summaries — in English and Kiswahili.

01**STEP 01**

Browser

 **User**

Uploads a PDF or pastes text directly in the browser.

**02****STEP 02**

Browser · On-device

 **PDF text extraction**

pdfjs-dist parses the PDF locally, on-device. The raw file never leaves the device.

pdfjs-dist

**Raw PDF never leaves the browser**

Only extracted text crosses this line — never the file. **The privacy boundary.**

**03****STEP 03**

Serverless

 **Serverless API**

Receives only the extracted text at `/api/analyse` on Vercel.

Vercel serverless



04

STEP 04

Model

🔮 LLM analysis

Google Gemini (free tier) with a strict prompt that forbids invention.

Google Gemini



05

STEP 05

LLM output

{ } Structured output

One JSON response — summary (EN/SW), key details, jargon terms, red flags with source citations, and next steps.



06

STEP 06

Frontend

🖥️ UI rendering

Renders the structured results — clean, bilingual, accessible.

Vite

React

TypeScript

Tailwind CSS

shadcn/ui



07

OUTPUTS

User

What the user sees

Everything above, in plain language — plus a mobile SMS preview.

Summary

Key details

Jargon buster

Red flags

Next steps

SMS preview

TRUST LAYER

Every claim carries a source citation.

“Show original text” reveals the exact source passage.

Wajibu · civic tech for clearer county procurement · built for Nyeri County tenders